

KEVIN KLEIN

Senior ML Engineer

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SUMMARY

Senior ML Engineer with 7+ years designing, deploying, and scaling ML and LLM systems in regulated, high-impact environments. Owned full ML lifecycle for a transformer-based clinical risk predictor (20K-patient cohort, recall >80%) and built a production clinical AI platform — RAG, tool-calling agents, multi-model serving — from zero to hospital deployment in 5 months, used daily by 50+ clinicians. Strong mathematical foundation (ETH Zürich, Magna Cum Laude).

SKILLS

LLM Systems vLLM · RAG pipelines · Agentic AI · Multi-agent orchestration · LangGraph · MCP · Qdrant · KV-cache optimisation
Core ML PyTorch · Transformers · HuggingFace · Ray · Time-series prediction · Imbalanced classification · Scikit-learn
Evals & Ops LLM eval frameworks · MLflow · DVC · Observability / tracing · Guardrails · Human-in-the-loop
Infra Kubernetes · Docker · GCP · Terraform · Argo CD · CI/CD · Linux

EXPERIENCE

Senior ML Engineer · X-cardiac

Berlin · 02/2022 – 02/2026

Early-stage cardiac AI startup; clinical decision support deployed across hospital cardiology workflows

Post-Op Cardiac Risk Predictor · Transformer · PyTorch · Ray · MLflow

- Owned the full ML development cycle (label design, architecture selection, distributed training, hospital deployment) for a transformer predicting postoperative complications (bleeding, acute renal failure) on a 20,000-patient cohort; recall >80%, precision 60–70%, enabling intervention up to 6h earlier.
- Implemented a published time-series transformer as the research baseline; drove key system design decisions (multi-label temporal offsets, focal loss for class imbalance, labelling strategy) that unlocked this recall where all prior configurations failed to generalise.
- Parallelized distributed training across 8-GPU cluster via Ray (~10× wallclock speedup), enabling 50+ Bayesian hyperparameter trials within 24h via Ray Tune; experiment reproducibility enforced via MLflow + DVC.

Clinical AI Platform · vLLM · RAG · Agentic AI · LLM Evals

- Built and deployed a production clinical LLM platform — RAG over a 500K-word clinical knowledge base, tool-calling agents over live patient data (FHIR, labs, vitals), and multi-model serving — zero to production in 5 months, supporting daily clinical decision-making across hospital cardiology workflows.
- Designed eval framework covering tool-call accuracy, retrieval relevance, and clinical safety; chose RAG over fine-tuning to preserve source auditability and support weekly knowledge updates without retraining — human-in-the-loop review for high-risk decisions and distributed tracing for multi-agent debugging.
- Achieved <300ms TTFT and 3,000+ tok/s across a multi-model stack (GPT-OSS 120B · Kimi 2.5 · Qwen Coder) on 8× H200 GPUs via vLLM — KV-cache tuning and continuous batching to sustain throughput under concurrent clinical load.

Data Scientist · 4flow

Berlin · 06/2019 – 06/2021

Document Object Detection · Object Detection · PyTorch

- Designed object detection model extracting structured data from logistics documents, reducing manual processing workload by 40%.

Email & Ticket NLP Pipeline · NLP · RNN · Multi-class Classification

- Drove a previously stalled NLP project to production leading 2 junior engineers: multi-class intent classifier routing 300 emails/day cut manual triage by 25%; RNN-based auto-fill eliminated data entry for ~50% of incoming tickets.

Risk Analytics Consultant · BDO Switzerland

Zürich · 10/2016 – 06/2018

- Led the firm's first international advisory engagement advising the European Central Bank on a major bank's asset quality review, and built a statistical loan selection algorithm cutting selection time by 50%.

EDUCATION

ETH Zurich · M.Sc. in Applied Mathematics

Zurich · 09/2014 – 06/2016

- Magna Cum Laude · Thesis: Order book models and price formation — stochastic modelling of high-frequency markets · Brownian Motion & Stochastic Calculus · Numerical Methods for SDEs/PDEs · Statistical Learning

Vienna University of Technology · B.Sc. in Technical Mathematics

Vienna · 10/2010 – 06/2014

LANGUAGES

German (Native) · English (Fluent) · Hebrew (Proficient)